



ENGINEERING WORKSHOP

FINAL PROJECT REPORT

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INTRODUCTION

Engineering workshops are crucial in the transition from theory to application. In electrical engineering, students must not only learn about the concepts of circuits but must also learn how to design, construct, analyze and test actual hardware circuits. Classroom teaching offers the basic knowledge and principles of electronics, while the practical projects in the workshop allow students to apply these fundamental concepts in various engineering projects. These types of projects provide students an understanding of the behavior of the electronic components in a real circuit, develop troubleshooting skills and teach students about hardware development and manufacturing. Engineering workshops are thus a vital tool for cultivating problem-solving ability, creativity, teamwork and technical self-confidence qualities that are key to the professional engineer.

A great advantage of engineering workshops is that they promote hands on experience. Students gain better understanding of tasks, such as circuit assembly, soldering, designing a PCB and drilling, wiring and testing an electronic system, when they physically do them. Students can interact with components and tools, allowing them to more clearly visualize engineering concepts and gain a deeper understanding of engineering. Workshops also help students to gain confidence in their technical skills since they learn to detect errors, troubleshoot faulty circuits and make improvements in their design. The hands on experience gained in workshop sessions is very useful as it helps students to prepare them for the future industrial work as well as professional engineering practices.

OBJECTIVE

In the design and implementation of an electronic oscillator circuit, we began with simulation in Proteus software and then progressed to creating the actual circuit. The simulation phase was also very important in determining the correct behavior of the circuit and making sure that the correct output waveform and frequency are being produced before proceeding to hardware implementation. At this stage, we studied different electronic components like resistors, capacitors, transistors and integrated circuits and understood their function in the circuit. The use of simulation software enabled us to detect and correct defects at an early stage, which helped to avoid hardware failure and reduce material wastage. This was a good time for us to reinforce our team knowledge of circuit analysis, waveform generation and hands on electronics.

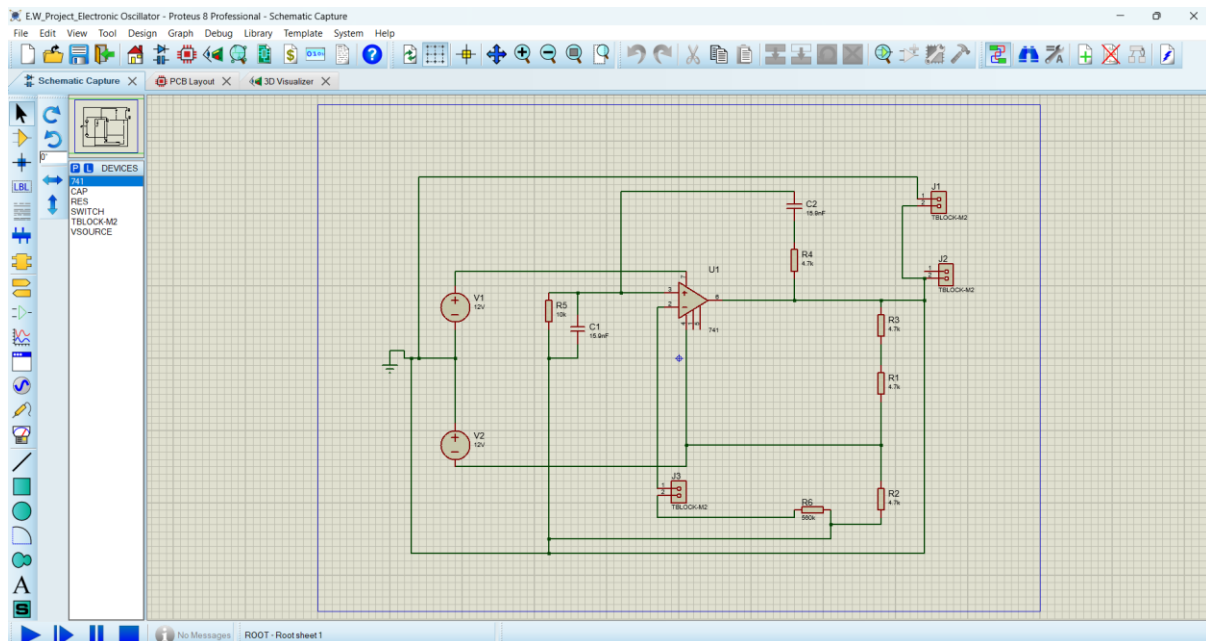
Once the simulation was done we got onto designing and making our PCB, which involved creating the layout, etching the board, drilling holes for the components and soldering the board. The PCB design is meticulously crafted to accommodate all components and optimize circuit layout. The PCB design is carefully planned to ensure that all components are properly placed and the circuit is designed efficiently. In practice, we encountered several challenges as we were making the PCB, including a few problems with the transfer to the PCB, that we

had to draw the tracks in with a permanent marker before we etched. These challenges enabled us to learn how to coordinate and adapt as a team as well as problem solving skills. In summary, this project gave us practical experience in circuit simulation, PCB fabrication and component assembly, as well as enhancing our collaborative, communication and technical skills. Assembling the oscillator circuit project provided us with good experience in the real engineering world and also in practical implementation.

PROCEDURE

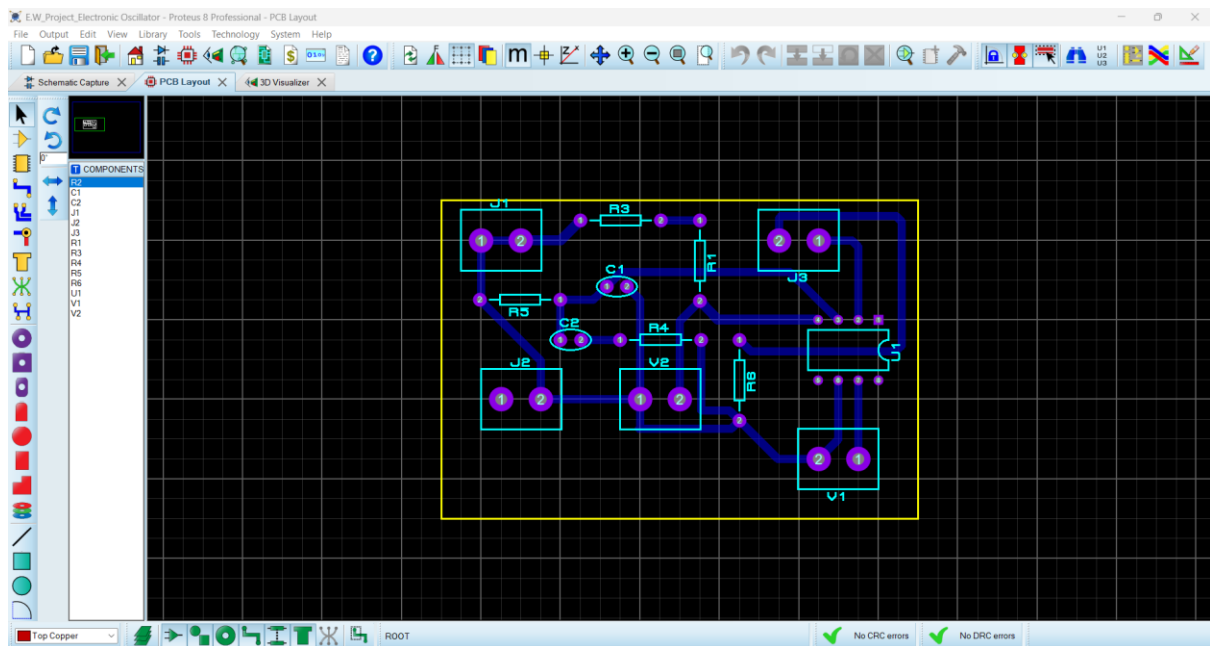
The schematic circuit of the electronic oscillator design was developed by first studying the design of the electronic oscillator and then determining the components of the electronic components that are required for the electronic oscillator. These components were selected with great care, depending on the frequency and output wave shape that is required from the circuit. All the components were then placed on the workspace and interconnected by drawing the virtual wires in Proteus software following the circuit diagram. Good grounding, input and output connections were provided to assure proper operation of the circuit.

Schematic Design:



Once the schematic circuit was completed, the PCB was designed using Proteus. At this point, all the electronic components were carefully positioned on the PCB workspace, ensuring a neat and efficient layout. The conductive tracks were then made between the parts to create the proper electrical connections as per the schematic. Special attention was paid to not have parts overlapping and to ensure that the current flows smoothly through the circuit. Easier and more efficient process for fabrication and soldering of the PCB layout. After completion, the layout was thoroughly reviewed for design issues or missing links and finalized for PCB printing and implementation.

PCB Design:

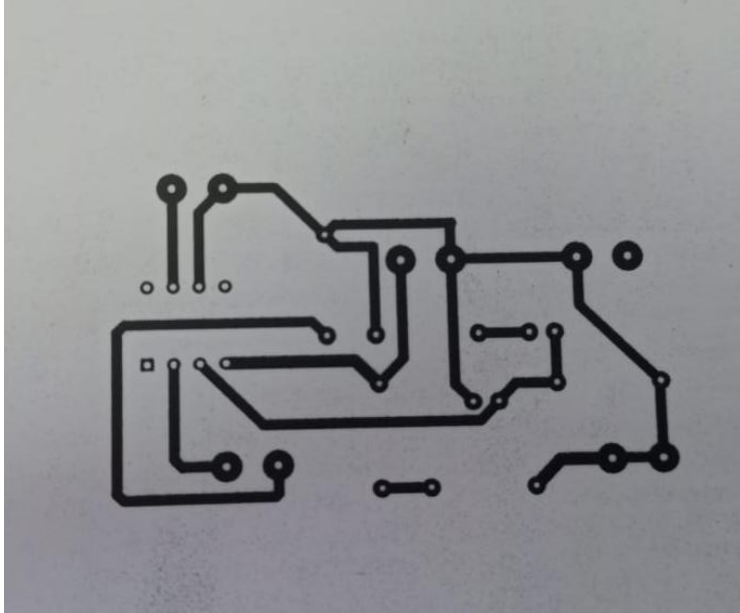


3D Visualization:



HARDWARE IMPLEMENTATION IMAGES:

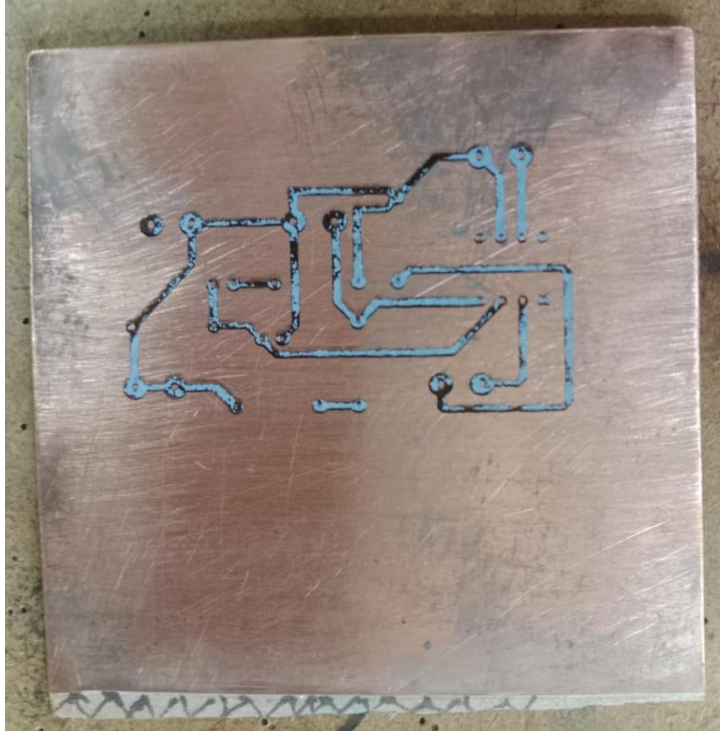
- ❖ **Printing the circuit design on glossy paper.**



- ❖ **Applying sand paper on PCB board after cutting it.**



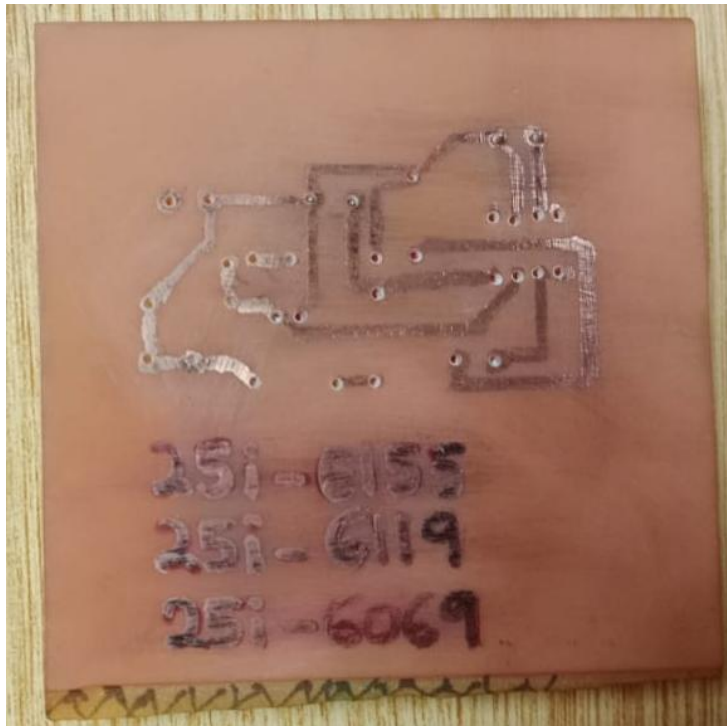
- ❖ After applying iron for 6 minutes with glossy sheet placed on PCB board, the print layout came on PCB.



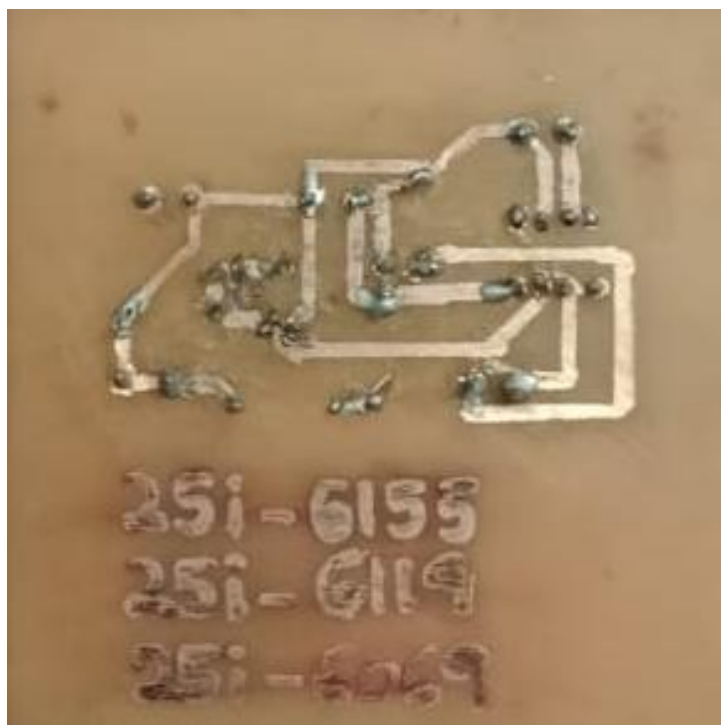
- ❖ The missing lines were traced by a permanent marker.



- ❖ After etching process, the PCB was drilled where the components were supposed to be soldered.



- ❖ The components were soldered with the copper on PCB board.



❖ Final PCB design.



RESULTS & OBSERVATIONS

It was difficult to determine the optimum size of the components footprints when designing the PCB layout. At the same time, it was not convenient to connect between the wires without shortening them. When we were attaching the T-block, it did cause a shortening but after consultation with our instructor we fixed that.

Once the layout of our printed circuit board was complete, we moved to etching. We did have a big issue however the circuit board's print was not fully transferred to the copper plate. Since this was not ideal, we had to finish the design by hand using a permanent marker, which worked out and let us continue. This was an essential step to make sure our circuit board was etched correctly and we could move on to putting the parts on the board. Even though it was a setback we properly etched the board, with a little extra work and some hands-on assistance.

We then printed a circuit board and drilled it for the components, according to the design. Because we used a marker to draw the layout it was a little bit tricky to find the exact position for the holes we had to drill, but by careful measurements we were able to get it just right and the holes ended up in the perfect position for our PCB.

We finally dipped our copper plate in ferric chloride (FeCl_3) to remove the excess copper from the plate. This method would only allow the copper to be on only those paths which the marker was protecting. This was crucial to our project as we feared we would over-etch the

copper plate. We waited a few minutes, then we removed the plate and finally we had a copper plate with a design that we wanted to assemble based on our PCB design.

Once the PCB was etched, the next step was to position all the required electronic components on the board and solder them onto it. This was an easy first step to take but soon, it was among the most important and complex elements in the entire process. This is because we had to take great care not to make any unnecessary connections or short circuits between the tracks, as this would spoil the entire board.

The circuit was being assembled and we did not try to place the components in the wrong places. We carefully followed the plan to get everything just right. Once the components were all together, we began to solder them together. We were still learning how to solder, though and had a little difficulty. Some parts were closer than usual making it easier for the solder to spread and join the wrong parts. Which it did while soldering, we had to take care not to get too much solder onto two paths that were in close proximity to each other, leading to a short circuit. We had to take care to not make this sort of error, as it can be complex to correct. Fortunately, it was possible to identify the issue and resolve it, but it was a valuable lesson on the need to tune the components when creating a circuit.

The problem was spotted immediately and remedied with careful removal of excess solder using a soldering iron, which again started to separate the tracks. This was a mistake that we corrected and the rest of the soldering went smoothly. All connections were made carefully and joints were both strong and clean. Next, using a wire cutter, we cut off the extra wire under the board. Eventually, we were able to assemble the PCB and the final board turned out as required with all parts in place and soldered properly.

CONCLUSION

Through this project we were able to get hands on experience with designing and making of PCB. We got to know about the significance of the correct sizing of the footprints, track routing and component placements to prevent short circuits while designing PCB. We had various problems during the fabrication, such as some points being not visible to drill, but we managed to overcome those with the help of a permanent marker; and some points were not transferred to the PCB, which we resolved with the permanent marker. This is the case of solder bridging that's we've encountered with soldering, where the components were not spaced apart enough, but it was corrected by cleaning up the excess solder. All these concerns rose our learning about making circuits. We also got an introduction to appropriate layering on proteus.

This project was very useful to us because it was a way to apply the information we learned in class to real world issues. We worked together, got to the bottom of difficult problems and got to know our technical skills as engineering students. That was a wonderful opportunity to learn, grow and to do that as a team.